

Skill Check 3: Building Better Code

Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Asking for help on specific questions. (It is fine to ask for clarification!)
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Skill Check Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully. For each two questions correct, you will receive one course point for a total of two course points.

Question 1: FUNCTIONS

Write a function that performs the calculation: $p = 3j^2 + 6kj + k^4 + 3$. Include an example function call to demonstrate the function in operation.

[illegible]

Question 2: LOOPS

Create a for loop that loops through and prints six colors of the rainbow.

[illegible]

Question 3: CONDITIONALS

You have a variable named `day` which holds the name of a day of the week. Write a switch statement that will print “Weekend!” if the day is Saturday or Sunday, “Friyay!” if the day is Friday, and “:(” if the day is a different weekday.

[illegible]

Question 4: COMBINATIONS

You have the following function `animal_sounds` which you can give a single animal name and get the sound back.

```
animal_sounds <- function(var){  
  if(var == "dog"){  
    sound <- "woof"  
  } else if (var == "cat"){  
    sound <- "meow"  
  } else if (var == "horse"){  
    sound <- "neigh"  
  } else {  
    stop("Please pick an animal sound I know!")  
  }  
  return(sound)  
}
```

Right now, the function is not vectorized, so to make all the animal sounds in the vector `animals`, a loop must use the function several times:

```
animals <- c("dog", "cat", "horse", "cat", "cat", "horse")  
for (x in animals) print(animal_sounds(x))
```

Convert the function `animal_sounds` so that it is vectorized, meaning that you can pass `animals` through it once (without a loop) and get back all the animal sounds in `animals` at once.

[illegible]

